

Budget Buddy App

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Track it. Own it. Win it....

Contents

1.Introduction.....	2
2. App Overview	2
2.1. App Name and Icon.....	2
3.Detailed Requirements List.....	4
4. User Interface Design	7
4.2. Navigation Diagram.....	8
5. Project Plan – Gantt Chart	9
5.1 Milestone Summary.....	10
5.2 Detailed Task Breakdown	10
6. Conclusion	12

1.Introduction

BudgetBuddy provides an off-line alternative to many other financial management apps for young adults in South Africa. The most prominent difference between BudgetBuddy and popular apps like Mint or YNAB is how BudgetBuddy stores data locally on your phone via a SQLite database. Users can therefore log-in to the app anywhere and at any time (if they have a device) regardless of whether they are connected to the internet.

2. App Overview

2.1. App Name and Icon

- **Name:** BudgetBuddy
- **Slogan** “Track it. Own it. Win it.”
- **Operation System** Android (React Native)
- **Target Market** Young adults (18–35)
- **Storage:** Local SQLite database (offline)
- **Price:** Free



Figure 1App icon

2.2. Description of Innovative Features

Innovative Feature	Description	Inspired By
Gamification	Budget Score (Bronze–Platinum), daily streaks, achievement badges.	Money Manager (badges)
Daily Streak Counter	A streak counter that keeps tracks of how many days the user has logged at least one expense. Streaks are gifted with milestone badges at 7, 30, and 100 days.	Money Manager (streaks)
Overspending Alerts	When a large expense is made, it will flash red.	Mint (budget alerts)
Receipt Photo Attachment	The app can also allow users to paste receipts for the expense and then display them along with the expense.	YNAB + Money Manager
Spending Trend Graph	Responsive graphs and side-by-side	Mint + Money Manager
Monthly Budget Progress Dashboard	Clear monthly progress bars per category.	YNAB (budget progress) + original
CSV Data Export	For tax records or deeper analysis.	Mint + YNAB

Table 1BudgetBuddy stands out through:

3.Detailed Requirements List

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BudgetBuddy requirements are grouped into nine sections. The **high-priority requirements** are crucial for launch; medium and low priorities will be seen as future enhancements.

Name	Requirement	Detailed Description & Acceptance Criteria	Priority
Authentication:	Register, login, logout, secure password storage.	Users register with a unique username and secure password (minimum 8 characters, hashed storage). They can log in, log out, and sessions persist until logout. Clear error messages are shown for invalid credentials.	High
Category Management	Default categories (Groceries, Transport, Entertainment, Utilities, Health) with ability to add/edit.	Default categories (Groceries, Transport, Entertainment, Utilities, Health) are provided. Users can create, edit, or delete categories, with optional color coding for charts. Deletion prompts a warning since linked expenses are affected.	High
Expense Entry	Add entries with amount, date, description, category, and optional photo.	Users add expenses with amount, date, description, and category. They can	High

	View and filter expenses by period.	attach one receipt photo, view it later, edit entries, or delete them permanently. Validation ensures correct input before saving.	
Budget Goals	Set Monthly Total Budget	Users set a total monthly budget and per-category limits. Overspending is detected and highlighted in red, with alerts triggered when limits are exceeded.	High
Expense List & Reporting	View All Expenses – User-Selectable Period	Expenses can be viewed by period (day, week, month, custom). Totals per category are displayed as lists or charts. Spending graphs show trends compared to previous periods. Data export to CSV is available for deeper analysis.	High
Data Storage	Local Database Storage	All data (users, categories, expenses, goals, photos) is stored locally in SQLite. Information persists across app restarts, and receipt photos are saved in private storage linked to entries.	High

Gamification	Budget Score	A daily Budget Score (0–100) is calculated based on staying within limits, logging consistency, and savings. Streak counters track consecutive days of logging, and badges reward milestones like first expense, streaks, or staying within budget.	Medium
Dashboard	Progress Dashboard	The home screen shows overall and per-category progress bars, Budget Score badge, streak count, and overspending highlights. A Quick Add button allows fast expense entry.	Medium
Settings & Profile	Settings Screen	Users can change username or password, toggle notifications, enable dark mode, export data, and log out. Notifications include streak milestones, overspending alerts, and reminders to log expenses.	Low

Table 2 Detailed Requirements List

4. User Interface Design

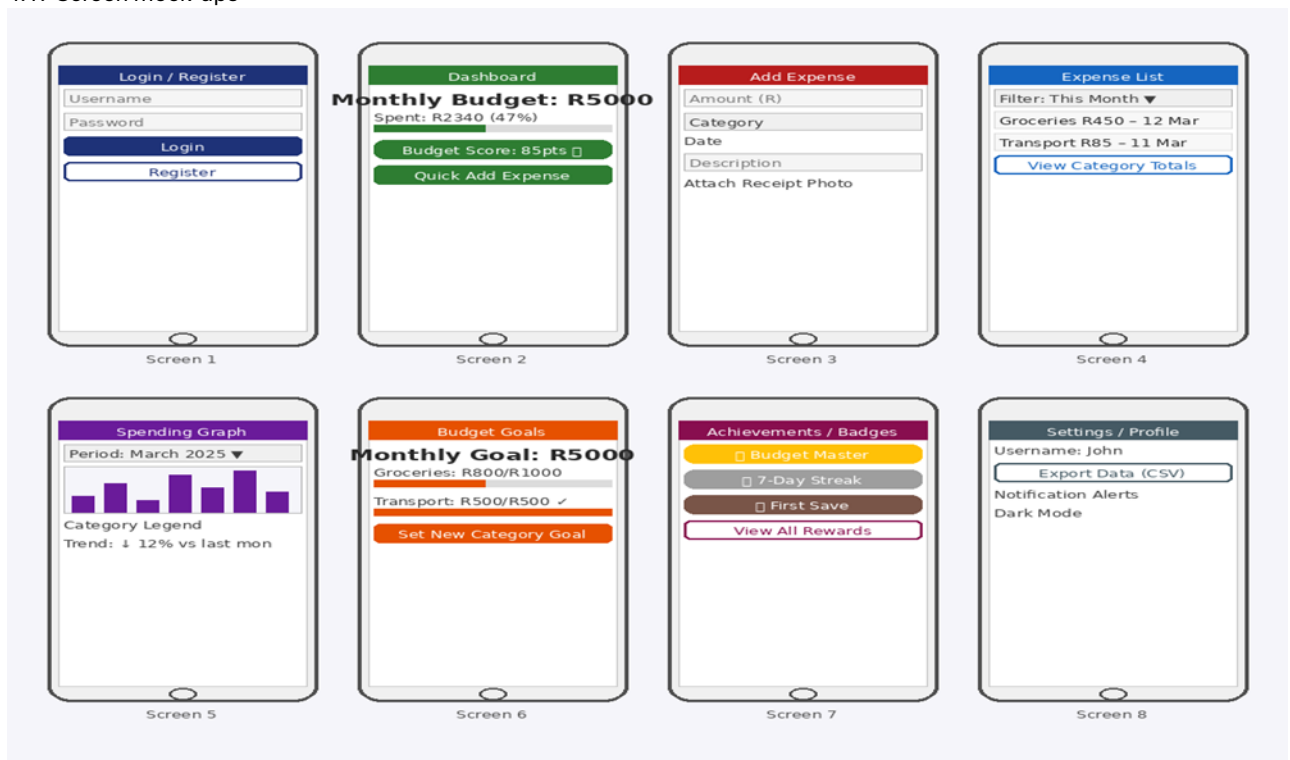
4. User Interface Design

BudgetBuddy will look clean using:

- **Colors:** Navy blue (main), forest green (savings/goals), red (overspending).
- **Screens (8 total):**
 1. Login/Register –secure entry point.
 2. Dashboard – hub with budget progress, score, streak, quick add.
 3. Add Expense – form with optional receipt photo.
 4. Expense List – filterable list with category totals.
 5. Spending Graph – bar chart with trends.
 6. Budget Goals – set limits, view overspending.
 7. Achievements – badges gallery.
 8. Settings – profile, notifications, dark mode, export.

Navigation is simple: the **Dashboard is the hub**, with bottom navigation to all other screens.

4.1. Screen mock-ups



. Figure 2. Screen Mock-upsEach

Screen No.	Screen Name	Purpose & Key Elements
Screen 1	Login / Register	Entry point. Secure entry point with username and password fields, plus option to register.
Screen 2	Dashboard (Home)	Central hub showing monthly budget progress, Budget Score, streaks, per-category bars, and Quick Add button.
Screen 3	Add Expense	Form to record expenses with amount, category, date, description, and optional receipt photo..
Screen 4	Expense List	Scrollable list of expenses filtered by period, showing details and linked receipt photos, with category totals.
Screen 5	Spending Graph	Bar chart of daily spending trends, color-coded by category, with comparison to previous period.
Screen 6	Budget Goals	List of categories with monthly limits and progress bars, highlighting overspending in red.
Screen 7	Achievements / Badges	Gallery of earned and unearned badges, with streak counter and milestone rewards.
Screen 8	Settings / Profile	Manage account details, change password, toggle notifications/dark mode, export data, and logout.

Table 3 Screen description and navigation

4.2. Navigation Diagram

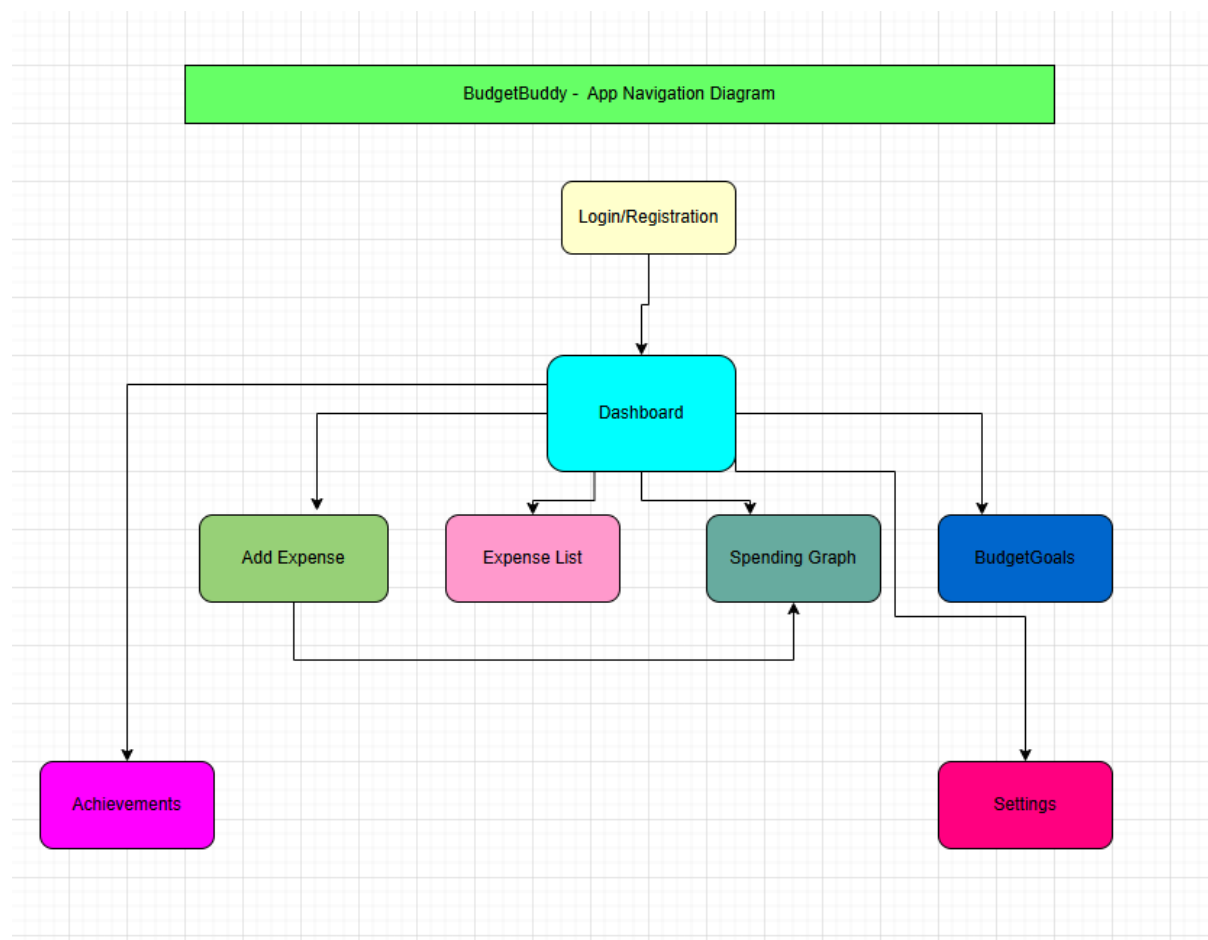


Figure 3 Navigation diagram

Navigation summary:

- Login → Dashboard
- Dashboard → All main features
- Add Expense → Expense List (after saving)
- All screens → Dashboard via navigation

5. Project Plan – Gantt Chart

5. Project Plan – Gantt Chart

The project runs over **7 weeks**, with clear milestones:

- **Weeks 1–2:** Research existing apps, finalize requirements and design.
- **Week 3:** Set up React Native project, implement authentication.
- **Weeks 4–5:** Build core features — expense entry, lists, budgets, dashboard.

- **Week 6:** Add advanced features — graphs, gamification, receipt photos.
- **Week 7:** we focused on testing, fixing bugs to improve the game, CSV export, and Notifications.

The deliverables would be the research documents, mock-ups, working app skeleton, final submission with all features according to the standards.

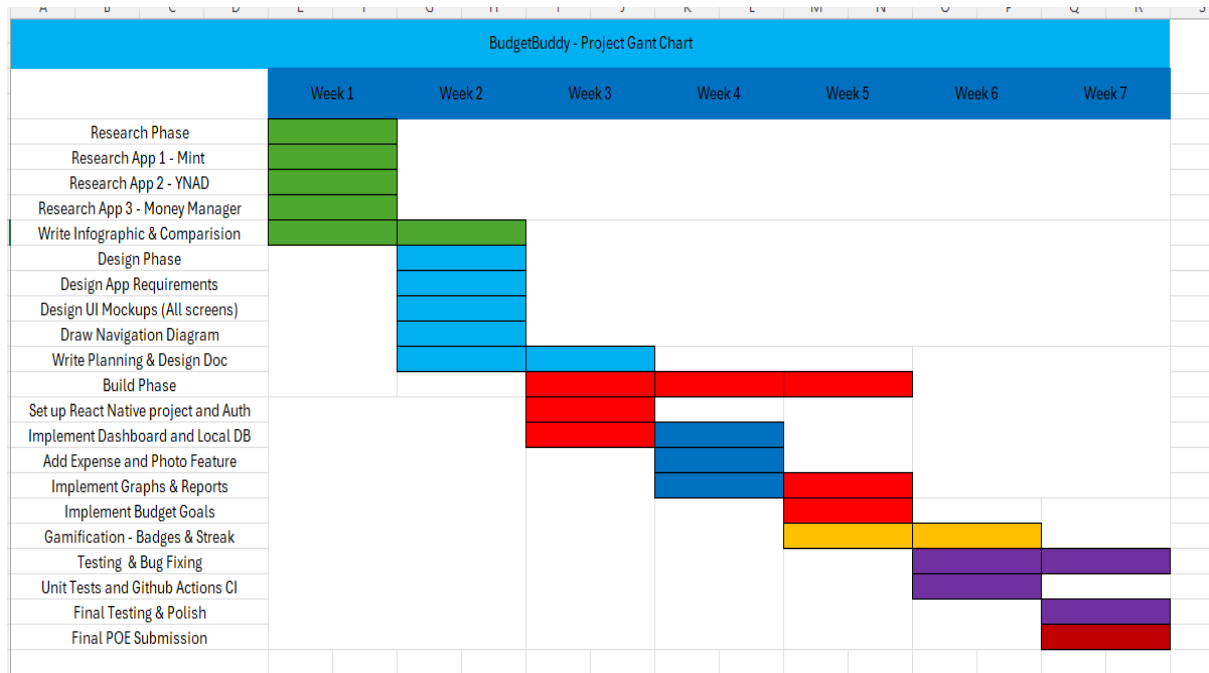


Figure 4 Gantt chart

5.1 Milestone Summary

Milestone

- Research completed (Week 2)
- Design completed (Week 3)
- Core features completed (Week 5)
- Advanced features completed (Week 6)
- Final submission (Week 7)

5.2 Detailed Task Breakdown

Week 1–2

- Research apps (Mint, YNAB, Money Manager)

- Identify strengths and features
- Create comparison and finalize research

Week 2–3

- Define requirements
- Design screens and navigation
- Create planning document

Week 3

- Set up React Native project
- Implement login and authentication

Week 4

- Build database structure
- Implement Add Expense feature
- Make an Expense List

Week 5

- Implement Budget Goals and Dashboard
- Add receipt photo functionality

Week 6

- Add graphs and gamification features
- Implement badges, score, and streaks
- Set up testing (CI and unit tests)

Week 7

- Perform full testing
- Fix bugs and improve UI
- Add CSV export and notifications
- Finalize and submit project

6. Conclusion

BudgetBuddy is an African application that exists to allow for budgeting in an easy manner. Through everyday use, customers will be able to check and analyse their spending, manage their budgets, and make great financial decisions. The application will include tracking, budgeting, reporting as basic features. Along with scoring, streaks, and badges for motivation as additional features. Budgeting will become more exciting to use with these features. The steps of developing an application is a carefully planned, with particular needs, design and development schedule over a 7-week period. This will make the application effective and user friendly with all app features working effectively

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